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Impact of High Hydrogen Operation on Combustor Performance

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Abstract

As the interest in high hydrogen operation is gaining momentum, this paper quantifies the impacts associated with switching from natural gas to 100% hydrogen, leveraging both modelling techniques and experimental data. From the modeling standpoint, a perfectly stirred reactor network model was setup in Cantera. Flame speed increases of up to 50 times that of natural gas were observed with increasing hydrogen content, indicating a significant increase in flashback propensity. This suggests that DLE combustion systems might offer an advantage over RQL systems, operating at low equivalence ratios where the flame speed impact is milder. Additionally, the model shows that the blow off time can be used to classify hydrogen operation into three regimes. With increasing hydrogen content, the BOT begins similar to that of propane and declines at different rates in each regime, establishing the added operational challenges associated with high hydrogen content operation. Equivalence ratio dependencies were investigated along with NOx penalties, where a predicted penalty of ~40-65% was observed within the flame temperature range of 1750-1950K. Experimentally, a new advanced mixer was used, enabling operation of the full spectrum of natural gas and hydrogen blends up to 100% hydrogen. The impact of hydrogen content on NOx emissions for a representative operating condition was investigated. Comparisons with the model predictions were made, revealing minor discrepancies, which were investigated and justified thru a mixedness and residence time framework. Finally, the authors show that the proper way to regulate future combustors running with 100% hydrogen should be based on NOx and not NOx15. The findings reported here help clarify and shape the future hydrogen enabling technologies, reaffirming the need for compact and shorter combustors than are used in current technologies.

Keywords

NOx emissions, NOx design curve, Hydrogen, Blow Off time, DLE, DLN, PFR, PSR, Emissions, Flashback, Auto-Ignition

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GE Global Research

Title: IMPACT OF HIGH HYDROGEN OPERATION ON COMBUSTOR PERFORMANCE

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IMPACT OF HIGH HYDROGEN OPERATION ON COMBUSTOR PERFORMANCE

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ABSTRACT

As the interest in high hydrogen operation is gaining momentum, this paper quantifies the impacts associated with switching from natural gas to 100% hydrogen, leveraging both modelling techniques and experimental data. From the modeling standpoint, a perfectly stirred reactor network model was setup in Cantera. Flame speed increases of up to 50 times that of natural gas were observed with increasing hydrogen content, indicating a significant increase in flashback propensity. This suggests that DLE combustion systems might offer an advantage over RQL systems, operating at low equivalence ratios where the flame speed impact is milder. Additionally, the model shows that the blow off time can be used to classify hydrogen operation into three regimes. With increasing hydrogen content, the BOT begins similar to that of propane and declines at different rates in each regime, establishing the added operational challenges associated with high hydrogen content operation. Equivalence ratio dependencies were investigated along with NOx penalties, where a predicted penalty of ~40-65% was observed within the flame temperature range of 1750-1950K. Experimentally, a new advanced mixer was used, enabling operation of the full spectrum of natural gas and hydrogen blends up to 100% hydrogen. The impact of hydrogen content on NOx emissions for a representative operating condition was investigated. Comparisons with the model predictions were made, revealing minor discrepancies, which were investigated and justified thru a mixedness and residence time framework. Finally, the authors show that the proper way to regulate future combustors running with 100% hydrogen should be based on NOx and not NOx15. The findings reported here help clarify and shape the future hydrogen enabling technologies, reaffirming the need for compact and shorter combustors than are used in current technologies.

1 NOMENCLATURE

P_3	=	Combustor feed pressure
T_3	=	Combustor inlet air and fuel temperature
TC	=	Thermocouple
$\Delta P/P$	=	Operating pressure drop across nozzle
NO_x	=	Nitrogen Oxides
CO	=	Carbon Monoxide
O_2	=	Oxygen
CO_2	=	Carbon Dioxide
H_2	=	Hydrogen
ppmvd	=	Particle per million volume dry
T_{flame}	=	Flame temperature-based on mixer air only
T_{Exit}	=	Combustor exit temperature (overall air/fuel)
ms	=	Combustor residence time in millisecond
p’	=	Thermo-acoustic instabilities pressure fluctuation in psi
pk-pk	=	Pressure fluctuations peak to peak
BOT	=	Blow off time
DLE	=	Dry Low Emissions combustion systems
EPA	=	Environmental protection agency
GER	=	General Electric Research
PSR	=	Perfectly stirred reactor
RQL	=	Rich-quench-lean combustion systems
( )15	=	Species corrected to 15% oxygen

2 INTRODUCTION

Hydrogen economy is currently being shaped and investigated as an enabler for a greener future. We read in industry the term “zero” associating any discussion on hydrogen as a potential fuel to be widely used in future. It is true that switching to 100% hydrogen means zero carbon dioxide and carbon monoxide, but not necessarily zero pollutant emissions. The primary remaining pollutant is NOx, where literature shows contradictory conclusions on the impact of hydrogen on NOx emissions. Some studies reported an increase, others reported a decrease, and some reported no change.

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Mohammad et al. [1] reviewed the different studies in the published literature. There work showed limited to no impact at up to 25% Hydrogen using an entitlement combustor (perfectly premixed). York *et al.* [2] reported a decreased of NOx emissions compared to natural gas operation. Luckerath *et al.* [3] reported a significant NOx increase with increase in hydrogen in the range of 20% -40%. However, there data shows a wide uncertainty in flame temperature estimation in addition to variation in air preheat temperature. Lantz et al. [4] reported an increase in NOx emissions but the burner was run at atmospheric conditions which limits the benefit of the data collected. Han S. Kim *et al.* [5] reported similar conclusions, but with a non-perfectly premixed system so the impoact of premixing is unknown. Peter Therklsen *et al.* [6] did a pathway study in an attempt to pinpoint the reason behind the NOx increase without conclusive results. Griebel *et al.* [7] reported that for a perfectly premixed case the NOx emissions are similar. Also, they showed that diffusion flame significantly results in higher NOx concentrations. Cozzi [8] reported similar results on a diffusion flame type combustor.

Combustion emissions are sensitive to almost all operating conditions. This paper is focused on NOx and CO, which are both sensitive to operating pressure, temperature, mixedness, residence time, flame temperature, diffusion portion of the flame, fuel composition and non-uniformities in the flow. Several authors documented the impact of pressure on combustion emissions [9-12]. The higher the flame temperature the higher the NO_x emissions penalty at elevated pressure. CO decreases with pressure. Residence time effect on NO_x emissions is documented in few publications, but was not thoroughly studied in a way that provides combustor designers to make the best selection based on target cycle conditions [6, 13, 14].

Perfectly premixed data are of extreme importance in quantifying gas turbine emissions as they isolate the impact of mixing from the data. This concept of entitlement data was introduced by Leonard and Correa [9] in the nineties. The system was simply a perforated flat pate burner running under perfectly premixed conditions. Elkady et. al [14] used a similar system to study the impact of operating conditions as well as to establish the most representative chemical reactor network model that can be used to model this flame . Mohammad et. al used the same system to show the impact of acoustics on the entitlement curve [15]. Also, Elkady et. al used it to study the impact of exhaust gas recirculation (EGR) on combustion emissions [16]. The rig data was also used for successful CO modeling close to lean blow out (LBO) [17, 18]. However, perfectly premixed data is not enough to quantify the changes that associates fuel changes in the field since the mixedness also changes. Fuel composition changes are typically associated with fuel jet momentum ratio changes which derives a change in mixing. Using perfectly premixed data to quantify the changes is not sufficient in this case as they don't capture the impact of mixedness.

This paper is focused on reporting the impact of switching from natural gas operation to 100% hydrogen operation on an actual industrial gas turbine DLE system. Recent studies have presented the results for blahblah. Here, two DLE mixers are used in a two-cup rig arrangement to quantify the result of switching from natural gas to 100% hydrogen on NOx and CO emissions. A reactor network perfectly premixed model is also used to quantify the impact and explain the experimental data. As high NOx were observed as the combustor was switched to hydrogen operation, a potential mitigation strategy is presented at the end of the paper. The mitigation for high NOx is suggested to be primarily through residence time reduction.

3 EXPERIMENTAL SETUP AND TEST CONDITIONS:

3.1 Test facility capability

The experimental data was acquired at the General Electric Research (GER) combustion test facility. This facility is used to develop future combustion technologies, enhance current technologies, as well as diagnose issues for systems in the field.

The test facility is comprised of three large air compressors, two gas compressors, a propane vaporizer, fuel blending station, bottle pack and trailer farm, boost air compressor and liquid fuel tank farm with different tanks that fit multiple liquid fuels including crude oil, JetA, Norpar, Diesel, Distillate and empty tanks for any liquid fuel of interest to be studied. The test facility is designed to enable testing for land based or aircraft engine-based combustion systems.

3.2 Test cell capability

An operational schematic of the test cell used for this study is illustrated in Figure 1. The main feed of natural gas can be mixed with other gaseous fuels. In this case, the natural gas stream was mixed with hydrogen from the fuel blending station. The mixing happens on the fly inside the test cell using a set of control valves and flow meters. The mixture is then routed to the combustor through different fuel lines. In this case, only two gaseous fuel circuits were required to run the system. The length of the tubes after mixing is larger than 200 L/D which ensures that the natural gas and hydrogen mixture is homogenous before it reaches the combustion space.

The test cell is capable of running liquid fuel and multiple air streams and there are several other gaseous fuel circuits, but they were not needed for this specific test.

3.3 Problem definition and combustor design

The combustion system utilized in this study is shown in Figure 2. The system is a two-cup based DLE combustion system. The cups are inserted in position through a heat shield assembly. The heat shield is cooled through impingement cooling. The flame tube is a bathtub shaped structure. It is back side cooled to limit the impact of secondary air on combustion emissions [19]. The liner back side cooling starts with a baffle that impinges air on the back of the liner and then the flow develops and flow in the sleeve to cool the liner aft portion down through convective cooling. The secondary air injector shown on the figure is sealed and not used for this study. The igniter on the side is a half inch hydrogen-based igniter that is water cooled. There are four emissions rakes installed at the combustor exit. Each rake has 5 sampling holes. Therefore, there are 20 sampling

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ports distributed on the combustor exit to enable high quality and spatially diverse emissions sampling.

The combustor structure here is maintained constant without any changes while varying the fuel from natural gas to hydrogen. The design is capable of handling any temperature spikes or shift in the heat transfer. It is important to freeze all these variables to isolate the quantified variables to only the variation of natural gas to hydrogen.

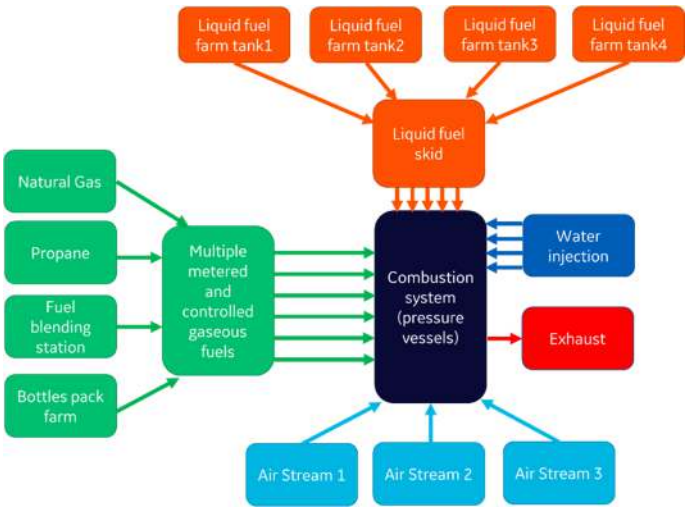

Figure 1. Test cell operational schematic layout

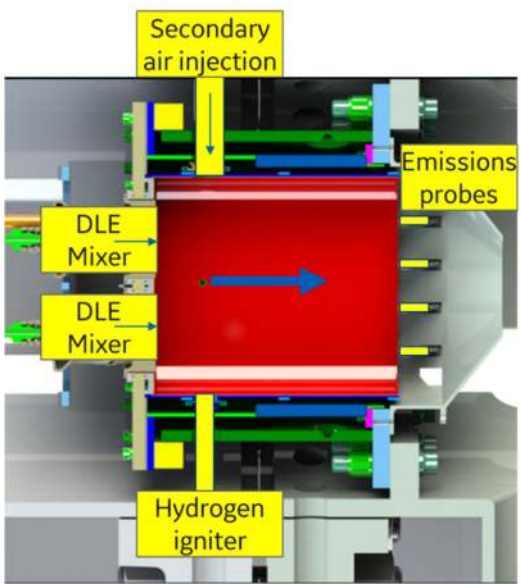

Figure 2. Combustor layout

3.4 Combustor instrumentation

The combustor is instrumented with thermocouples and pressure measurements points to ensure that all parameters of interest are running within the operating range and acceptable operating conditions.

There are several thermocouples on the liners to ensure that the metal temperature does not exceed the acceptable operating temperatures for the liner material, Hast Alloy X in this case. Also, the heat shield is instrumented with several thermocouples for the same reason.

There are 4 pressure measurement points on the combustor and two quarter inch tubes used for monitoring static and dynamic pressures to ensure that the amplitude of combustion acoustics will not result in mechanical failure. Typically, a limit is set for the high frequency dynamics (1000-6000 Hz) at 5 psi pk-pk and for the low frequency dynamics (0-1000 Hz) at 12 psi pk-pk.

The four static pressure measurement points provide an average pressure in the combustion zone that is used to estimate the amount of air flowing in each mixer, based on the mixer pressure drop. Based on these measurement values and the amount of fuel flow prescribed, the fuel-air ratio (FAR) can be estimated, and hence the flame temperature for each mixer.

The metered air, metered fuel, estimated mixer air, and emissions, specifically the oxygen and carbon dioxide balance, are used to calculate multiple closures that are used to assess the integrity and quality of the test. The main quality metrics are air closure, fuel closure, oxygen closure and carbon dioxide closure. Also, methane closure is used before light up to ensure that the light up will be safe and within expectations.

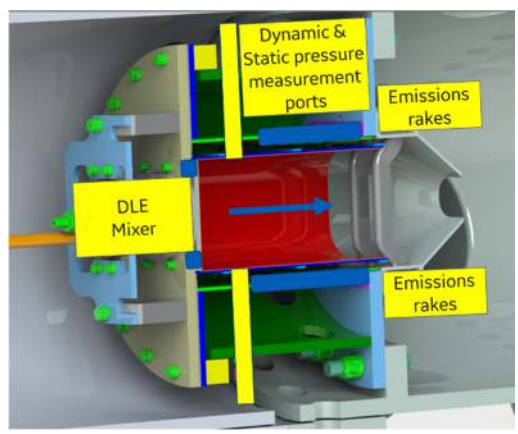

Figure 3. Flow path, static and dynamic pressure measurement locations

3.5 Emissions sampling system

The emissions samples collected thru the rake sampling holes flow through an electrically heated hose that help maintain the sample temperature at 422K. This temperature ensures no condensation of the water in the sample gas will occur before feeding the sample to the analyzer. This is important as NO_2 is water soluble. If water condensation occurs, then NO_2 may be lost and hence the total NO_x measured will be in error. The Environmental Protection Agency (EPA) method 7E is of extreme importance to be referred to by researchers and industrial entities performing emissions sampling [20].

The emissions stand is composed of a series of analyzers that are used to measure all major components of the sampled gas stream. Mainly we are interested in oxygen (O_2), nitric

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oxides (NO_x), carbon monoxide (CO), unburned hydrocarbons (UHC), methane (CH_4) and carbon dioxide (CO_2). NO_x15 refers to the standard correction of the NO_x emissions to 15% oxygen. The same terminology applies for CO.

3.6 Operating conditions, data storage and reporting

This study, both modeling and experimentally, is performed at representative gas turbine operating conditions summarized in Table 1. The target residence time is ~ 8 ms at the design point operating conditions. An increase of the pressure drop to 8.75% changes the residence time to roughly 5.6 ms, information which will be useful in Section 4.5. The fuel is pumped to the combustion system without pre-heating. There are two mixers installed in the system and both were operated at identical conditions (no burner bias for this study). That is the global FAR per mixer was the same and is equal to the system global FAR. Reported data is logged at a rate of 1 Hz. The data presented here is collected at steady state where each point is an average of at least 10 seconds worth of data.

Table 1. Baseline Operating Conditions

Parameter	Value
P3	23 atm
T3	785 K
ΔP/P	4%

4 RESULTS AND DISCUSSION

4.1 Impact of 100% hydrogen operation on flame speed and Blow Off Time

The laminar flame speed (S_L) is of prime importance in understanding the propensity of DLE mixers to flashback. In addition, it dictates the flame area and overall length as it describes the consumption rate of the air and fuel mixture. However, the flow is primarily turbulent in practically all industrial combustion systems, therefore a useful quantity is the turbulent flame speed (S_T) which is typically correlated to S_L. One sample formula is given in Eq.(1) where U' is the fluctuating component of the velocity [21]. The ratio can be between 2-50 depending on the problem being solved.

$$S_T/S_L \sim\left[1+\left(\overline{U'^2}/S_L^2\right)\right]^{1/2} \quad (1)$$

To investigate these trends, a perfectly stirred reactor (PSR) network model was setup in Cantera and exercised at the operating conditions listed in Table 1. The PSR residence time was set to 0.5ms. Figure 4 shows the variation of the laminar flame speed with equivalence ratio for methane and hydrogen flames. An alternate representation, in terms of ratio of hydrogen laminar flame speed to methane laminar flame speed, is shown on the secondary axis. A major challenge of burning hydrogen is apparent from the figure as the hydrogen flame speed velocity is orders of magnitude larger than methane. As equivalence ratio increases so does this flame speed ratio, representing increased sensitivity and susceptibility to flash back. This suggests that leaner combustion systems would have an advantage when switching to burning hydrogen, having more modest flash back characteristics. For example, DLE combustion systems, typically operating at equivalence ratios less than 0.75, will have fewer hydrogen related flash back issues than diffusion based or RQL combustion systems, which operate at the larger equivalence ratios. In addition, combustor part durability is also a concern with increasing flame speed as the flame is expected to sit much closer to the hardware or attachment surfaces. Thus, this is one of the few cases where lean burn systems burning hydrogen can be more advantageous when compared to rich burn systems.

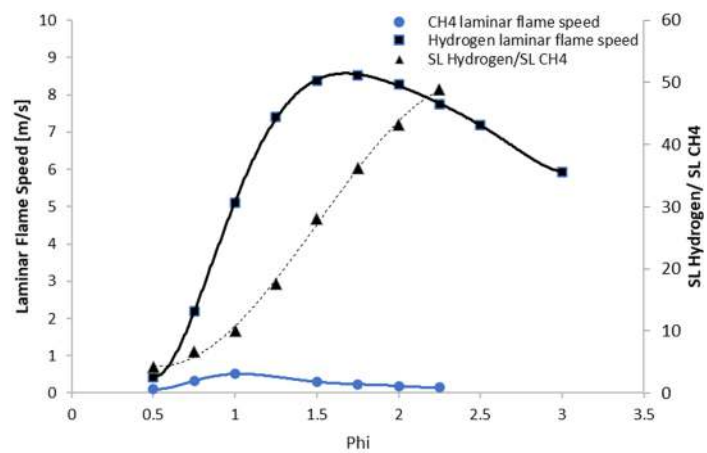

Figure 4. Laminar flame speed of hydrogen (H2) and Methane (CH4) vs equivalence ratio (Phi)

Gas turbine manufacturers typically test mixers in a controlled environment at representative operating conditions of the machine for flame holding. These tests confirm the ability of the mixer to both stabilize a flame in the designed reaction zone and to clear a flame that is induced into the mixer, irrespective of the reason it was induced. Field experience has shown that this is a key performance parameter that can result in millions of dollars in losses if not tackled properly. Blow off time (BOT), defined as the minimum PSR residence time required for the fuel air mixture to have reaction or heat release, can be used to numerically confirm that the mixer will be resistant to flame holding events. Blouch *et. al* showed that for propane and ethane operation, BOT was a very accurate indicator of flame holding acceptability limits for GE LM2500 and LM6000 machines using DLE1.5 mixers technology [22]. If the BOT is higher than the residence time in the mixer then there is no risk of flame holding, and if BOT is lower than the mixer residence time then flashback or flame holding becomes an issue.

The PSR network model was once again leveraged to investigate BOT trends for various fuel blends at the operating conditions listed in Table 1. The equivalence ratio of interest is assumed to be unity to represent flame holding events close to the fuel injection orifices inside the mixer. The model was run for various natural gas blends of propane (serving as a baseline) and hydrogen ranging from 10% to 100%. Figure 5 shows a

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direct comparison of the BOT for these concentration sweeps. The extremely low values with significant hydrogen concentration indicate large flame holding and flashback risk, being yet another challenge with high hydrogen content combustion. For example, the BOT at 100% propane is equivalent to that at 43% hydrogen (roughly 3.5 micro-seconds). Propane was chosen as the baseline for comparison as it is known to be already sensitive to flashback and flame holding. The data shows that at 80% hydrogen the BOT is roughly 1 micro-second, establishing an extremely aggressive requirement. However, up to as high as 30% hydrogen operation there is no significant difference between propane and hydrogen trends, suggesting that the operation of the combustor free of flame holding or flashback events should be relatively easily achieved in this range.

Based on these curves, three operational regimes can be defined. First is the easiest which is up to 30% concentrations, where the trends are almost identical to that of propane. Next, is between 30-80% where both concentration increases lead to decreasing BOT values, albeit at different rates. Finally, at concentration blends higher than 80%, propane seems to level off while hydrogen continues to decline, indicating increasingly challenging operating requirements.

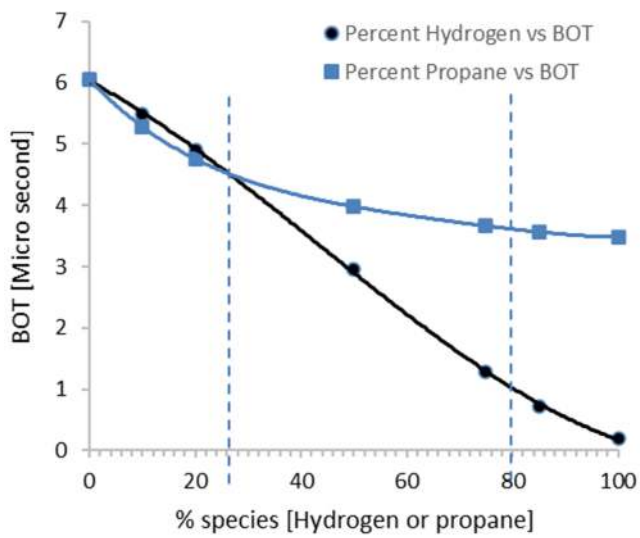

Figure 5. Hydrogen vs Propane BOT sweeps, where vertical dashed lines denote the regime boundaries

0 shows the sensitivity of BOT to operating pressure and temperature variations well within the realm of possibility for fielded systems. The data shows that the air preheat temperature has a very strong effect on reducing the BOT, and therefore engines running at higher pressure ratios have an added level of difficulty. This becomes an increasing concern as our technology is driving towards higher pressure ratios as we strive for higher efficiencies. The impact of higher operating pressure was much more minor.

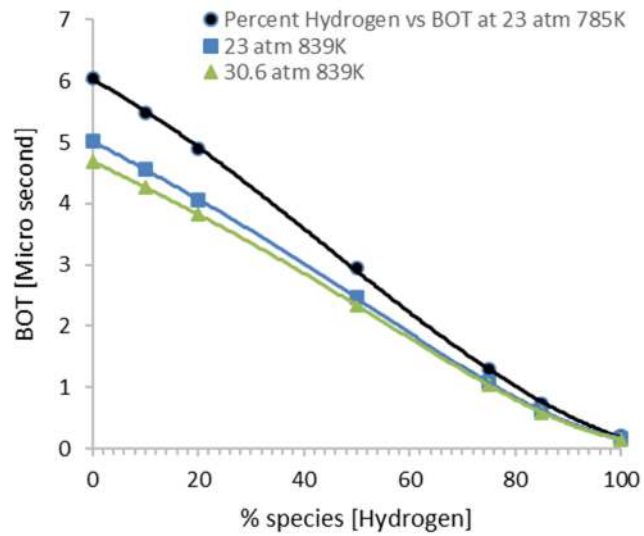

Figure 6. Sensitivity of BOT to operating pressure and temperature for different hydrogen blends

Figure 7 shows the variation of BOT with equivalence ratio for a 100% hydrogen air mixture running at ~23 atm and 785K. In the previous concentration sweep studies, it was mentioned that the PSR network model was set to an equivalence ratio of unity. This curve shows that the BOT only recovers to the value observed for pure propane at equivalence ratio values lower than 0.42 or higher than 6. Within the range 0.47-1.5 the BOT is extremely low (less than 1 micro-seconds). This indicates that hydrogen combustion systems running under partially premixed or close to perfectly mixed conditions will require additional hardware design features to avoid flame holding, rather than relying on chemistry as is standard with traditional fuels. The work of York *et al*. shows one such example of a system redesigned to be able to operate at 100% hydrogen. However, operating pressures and temperatures are also an important factor in the performance evaluation of systems intended to operate under 100% hydrogen [2].

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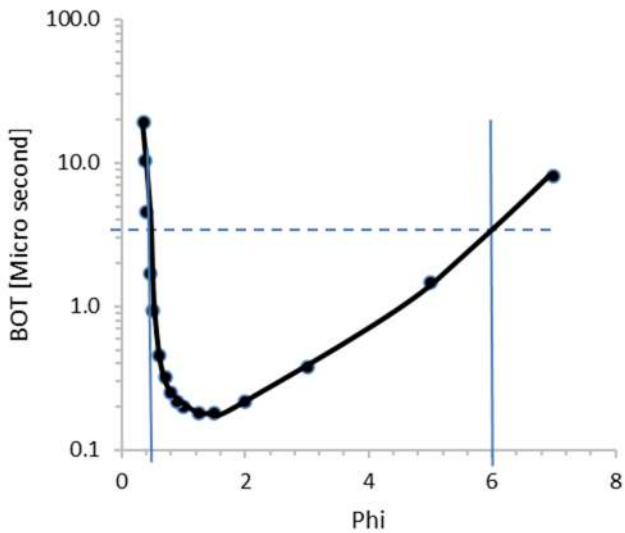

Figure 7. Impact of equivalence ratio (Phi) on the BOT for 100% hydrogen at 23 atm and 785K.

4.2 Impact of 100% hydrogen operation on NOx emissions using chemical kinetics

Previous model results shown so far in this work have leveraged a PSR network model, however, to investigate the chemistry of these simplified combustion systems now a joint PSR and plug flow reactor (PFR) network model will be leveraged. This is a common practice and will be denoted PSR-PFR in this work. Elkady *et al*. used a similar network model to predict the emissions of a perforated entitlement burner running with natural gas (perfectly premixed) [23] and achieved very good agreement with experimental data. The sensitivity of the model was evaluated using different reaction mechanisms and at different operating conditions, having widespread agreement with the experimental data. Mohammad *et al*. [15] used the same version of this combustion system to study the impact of hydrogen enrichment, however only up to 25% hydrogen by volume and showing very limited impact on NOx emissions.

Figure 8 schematically depicts the PSR-PFR network model constructed and leveraged to compare predicted emissions under natural gas and hydrogen operation. Two different chemical mechanisms were initially investigated; an industry standard, GRI3.0, and a more aggressive chemical detailing, MFC2. This was done to gain understanding into the sensitivity of hydrogen content combustion metrics on the mechanism used. Figure 9 shows the laminar flame speed computed for each mechanism across hydrogen content sweeps. It can be observed that the flame speed is estimated to increase between 2-6 times as the flame transitions from natural gas to 100% hydrogen content (at this specific flame temperature. Figure 4 shows the general trend with equivalence ratio). Aside from Figure 9, all subsequent results for laminar flame speed throughout this paper will utilize the GRI3.0 mechanism as it is the current industrial standard.

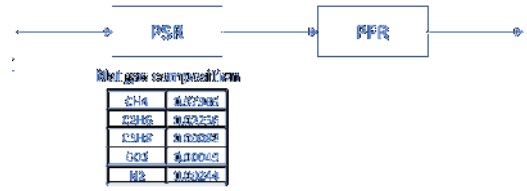

Figure 8. Schematic of the PSR-PFR network model ran at baseline conditions (See Table 1)

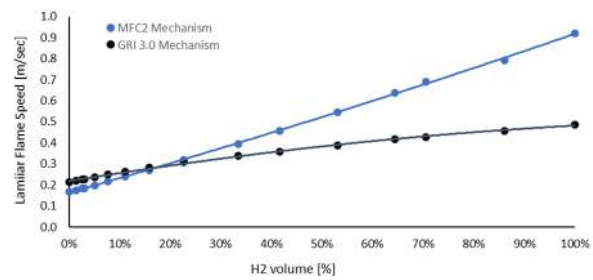

Figure 9. Laminar flame speed vs hydrogen content by volume for different chemical mechanisms at Tflame ~1950K and baseline conditions (Table 1)

Along with laminar flame speed, another important parameter in the network model are the residence times. To assess the impact of the PSR-PFR residence time split, the PSR size is assumed to inversely change with the laminar flame speed squared. This results from the chemical time scale, τ_{CTS}, being inversely proportional to the square of the laminar flame speed as follows:

$$\tau_{CTS} = \frac{\left(\lambda/_{Cp}\right)_{To}}{\rho_u S_L^2} \quad (2)$$

For the subsequent model results shown, the residence time split between the PSR and PFR was kept equal, at 0.5ms/7.5 ms each (for an overall system residence time modeled of 8 ms). This was justified with accompanying numerical investigations which showed the impact of the PSR residence time on NOx emissions to be so minor that it was essentially negligible. For example, it was observed that decreasing the size of the PSR, for hydrogen operation, to 20% that of the natural gas operation resulted in roughly a 1% reduction in NOx. Thus, for simplicity they were kept equal.

Figure 10 shows the PSR-PFR network model results, comparing NOx values for natural gas and hydrogen flames. These curves show the typical exponential NOx growth, as is common and expected, versus flame temperature (denoted *TFlame* in the figure). A significant NOx penalty can be observed

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as the fuel is switched from natural gas to hydrogen. This penalty becomes more obvious when the ratio of NOx is plotted, and thus NOx_{H2}/NOx_{CH4} is also shown in Figure 10. Most modern DLE combustion systems operate with flame temperatures in the range of 1750-1950 K, corresponding to a model predicted NOx penalty in the range of ~65-40%. Also, it is observed that as the temperature increases the percentage penalty decreases. This effect might be attributed to the decreases in prompt NOx as temperature increases, resulting in decreasing flame lengths.

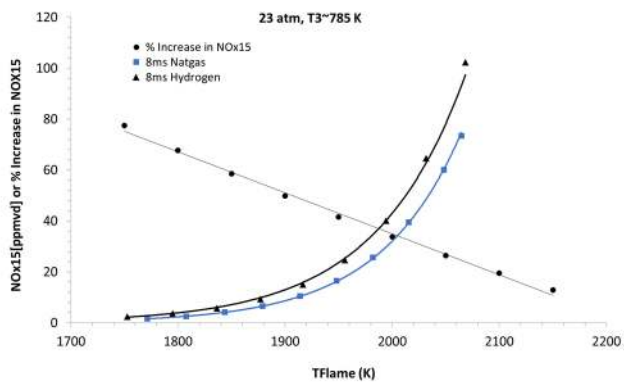

Figure 10. Model prediction of hydrogen vs natural gas NOx15 operation for a 23.8 atm, 783.1K and 8 ms

4.3 Impact of 100% hydrogen operation on DLE combustion systems

Having gained some preliminary insight into the impact of hydrogen content on NOx emissions from the network model framework, the next step is to investigate the experimental results. The combustion system shown in Figure 2 was used to experimentally quantify the impact of hydrogen on emissions thru various hydrogen content sweeps; specifically the system was run at hydrogen percentages of 30, 40, 50, 60, 70 and 100% with flame temperature of 1935 K. Additionally, two temperature sweeps were conducted at 50% and 100% hydrogen in order to study the impact of operating temperature on NOx growth. These were performed by varying the flame temperature while keeping everything else the same and collecting data, resulting in curves commonly known as NOx design curves. These NOx design curves provide combustor designers with critical information required to target and achieve emissions requirements and thus are highly sought after in industry.

As a brief aside, it is important to recognize that the FAA NOx and CO regulations actually regulate NOx15 and CO15, rather than NOx and CO. These modified emissions metrics represent the respective species emissions corrected to 15% oxygen using the following formula:

$$\boldsymbol{X15} = \frac{\mathbf{20.9-15}}{\mathbf{20.9-O2}\ [\%]} * \boldsymbol{X}\ [\boldsymbol{ppm}] \qquad (3)$$

where X denotes the species concentration in ppm and O2 is the nominal oxygen percentage in the original measurement. Therefore, often results will be presented against and discussed with respect to this modified metric.

Figure 11 shows the three NOx design curves obtained, one for natural gas and the two hydrogen sweeps at 50% and 100% respectively. It can be observed that as the hydrogen content increases, the NOx emissions increase significantly following a second-degree polynomial. As a note, this figure shows the design curves plotted against TO2, being the temperature measured at the exit of the combustor. In this work we will utilize Tflame, obtained from model results, and TO2, measured from experiments, interchangeably.

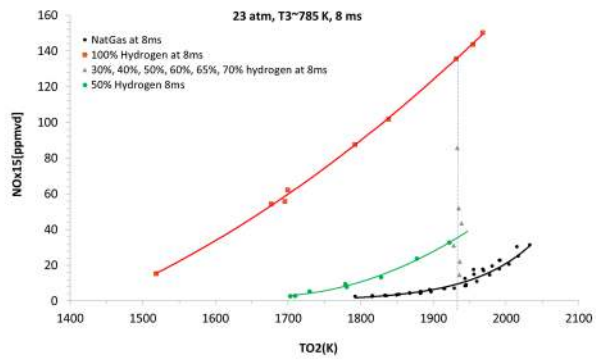

Figure 11. NOX15 Emissions data for natural gas and hydrogen. Shown are three NOx design curves and a hydrogen content sweep at 1935K.

Figure 12 shows the hydrogen content sweep, which is also denoted on Figure 11 by the dashed line, establishing a transfer function for the impact of hydrogen on NOx emissions. Using the regimes defined in Figure 5 for BOT, it can be observed that at low hydrogen percentages there is very limited impact of increasing hydrogen content on NOx. Above this regime however, the impact is significant. In the middle regime, the increase in NOx with hydrogen content seems exponential, while in the final high content regime the increase appears linear.

By referring to Figure 10, comparisons between the network modeling framework and experimental data can be made. Extracting the values at 1935 K, the model shows a predicted NOx increase of ~45%, being equivalent to the 100% hydrogen content point in Figure 12, while the experimental data shows an increase of ~1200%. This massive difference can be attributed to the hydrogen NOx prediction from the model being well below the experimentally observed values. Notice how the natural gas prediction seems reasonable, while the hydrogen prediction is almost an order of magnitude off. Physically, this can be attributed to the model inability to capture the variation

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in mixing or the flame-flow interaction impact that is associated with hydrogen addition, effects that will be further explored next.

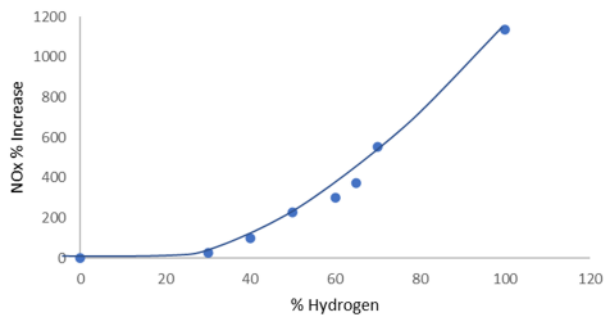

Figure 12. Impact of hydrogen addition on NOx15 emissions measured experimentally

Although NOx is the primary emission discussed when referring to pure hydrogen combustion due to the absence of carbon atoms, however, when dealing with hydrogen blends CO is still a concern requiring attention. Figure 13 shows the impact of hydrogen content (blended with natural gas) on CO. The observed absence of CO at 100% hydrogen was expected and serves as mental grounding, however, at 50% hydrogen the trend is very interesting. With 50% hydrogen, there is just 30 ppm CO15 at 1700K which corresponds to 465K of turn down capability. The elevated levels of NOx and significant reduction in CO points us towards the solution to enable 100% hydrogen with emissions compliance. It is known that CO levels decrease with longer residence times, however, due to the large margin here with the 50% blend (or complete absence of CO with 100% hydrogen), this points towards using shortened combustors or combustors with faster residence times as to limit the amount of NOx formation.

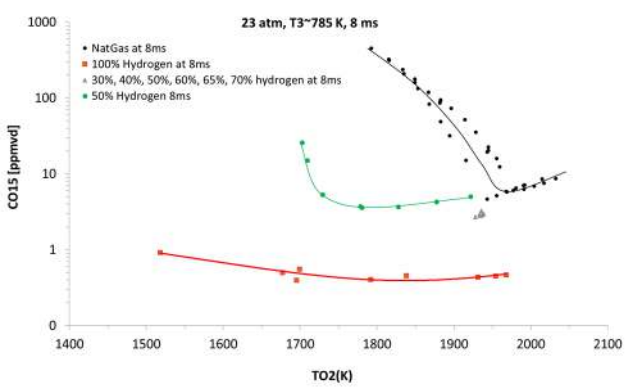

Figure 13. Impact of hydrogen addition on CO15 at 8.5 ms

As a final point on these experimental emissions trends, upon close inspection of Figure 11 and Figure 13, it can be observed that for the natural gas data there are actually two sets of results plotted. Low and high emission branches exist for CO and NOx at low and high temperature values, respectively. For each of these dataset pairs, the same data was acquired, however the temperature sweeps were done from different sides, one increasing and one decreasing. The resulting behavior is the hysteresis phenomena associated with the flame shape changes as the temperature is approached from the lower side or the higher side (also commonly known as bifurcations or bi modal behavior in combustion). It can be seen that the bifurcation point, i.e. temperature, appears to be around 3025F.

*Add equivalent of Figure 12 but for CO?

4.4 Incorporating effect of un-mixedness in chemical kinetics model

Published research has shown that the PSR-PFR reactor network model approach is able to capture, with high accuracy, the NOx emissions of simple natural gas jet flames that are perfectly premixed [14]. From Figure 10 it can be observed that at a flame temperature of 3025F, the reactor network predicts NOx15 to be 12.5 ppm under natural gas operation, while experimental data shows emissions values between 8.5pm-12ppm (depending on hysteresis direction or flame shape). This implies that the system with natural gas is running in an operational mode close to perfectly premixed.

Recall the previous discission regarding the large deviations between network model NOx predictions and experimental results. The jet momentum ratio of the pure hydrogen fuel, relative to pure natural gas, is 1.25, suggesting that a significant impact on mixing is to be expected when switching from natural gas to hydrogen operation. As a result of this higher fluid momentum, the mixing profile likely becomes more center peaked with hydrogen. While this may help with flash back and flame holding, it will also drive higher NOx emissions due to blahblah.

To capture this impact and remedy some of the model deficiencies, we propose a simple framework to estimate and incorporate the un-mixedness effect into the modelling results. The reactor network model can be used to predict NOx for any equivalence ratio of interest if it is assumed to be uniform across the flow. Thus, for the proposed framework, we will assume that the NOx emissions for the combustion system shown in Figure 2 are a mixture of perfectly premixed emissions and a hot spot, representing a localized region of high fuel content, operating at an equivalence ratio of unity; chosen as this represents the worst-case scenario with maximum adiabatic flame temperature. At the baseline condition, now assuming an equivalence ration of unity, the model predicts NOx values of roughly ~3733.3 ppm while burning pure hydrogen. Experimentally, there is 259.7 ppm NOx generated at the same conditions of interest where the model predicts 43 ppm. The split is roughly 5.75%. Therefore, the deterioration of un-mixedness as we switch from natural gas to hydrogen can be quantified at roughly 6%.

This framework assumes that the kinetics model fully captures the proper physics and that deviations can exclusively be attributed to un-mixedness that takes place as a result of the burner design not being intended for 100% hydrogen operation. Another important factor that could contribute is the residence

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time impact in the post flame zone. One method to confirm this assumption would be to have dynamic fuel orifice holes such that the momentum ratio is constant with various hydrogen content blends; enlarged holes for higher hydrogen contents. However, these hardware modifications and test reruns are beyond the scope of the current work. This work is focused on simply quantifying the changes that associate switching from natural gas operation to hydrogen content blends, up to 100% hydrogen. The results confirm that NOx emissions will increase significantly either way. As was shown by Figure 12, NOx15 can increase as much as 1200% when switching from natural gas to pure hydrogen operation. Restoring the mixedness will help reduce the impact but will never eliminate it fully. In the next section a simple method will be used to show how to reduce NOx15 without hardware changes.

4.5 Abatement of NOx15 via pressure drop (residence time decrease)

The increase in NOx levels associated with moderate to high hydrogen concentrations is of concern. An attempt to reduce the NOx can be made by increasing the pressure drop, i.e. reducing the residence time, so that thermal NOx has less time to form. However, along with negatively impacting CO for carbon containing fuels, reductions in the residence time also impacts the degree of mixing. Unlike the jet momentum theory, investigating this effect does not require hardware modifications, only an operational condition change. Therefore, another round of experimental data was obtained, for both natural gas and hydrogen content and temperature sweeps, at a larger pressure drop of 8.75%, i.e. reduced residence time. Recall, as was denoted in Table 1, Figure 11 and Figure 13 displayed results for the design point with a pressure drop of 4%. This elevated pressure drop value corresponds to a reactor residence time of 5.6ms, which is roughly a 2.5ms reduction from the baseline case.

Figure 14 and Figure 15 show the NOx and CO emissions trends for natural gas, hydrogen, and, various representative hydrogen content blends at the elevated pressure drop. As was expected, compared to the baseline conditions the NOx values are significantly lower overall. Additionally, for the partial hydrogen blends and natural gas data, the CO emissions values are higher. However, interestingly the ratio of NOx increase associated with switching from natural gas to pure hydrogen operation is roughly the same under both pressure drops which indicates limited impact on mixing in this case. Figure 16 can be used to estimate the impact of residence time under 100% hydrogen operation. The impact is huge. It is roughly 20 ppm / ms. That is for every ms reduction in residence time there is an equivalent reduction in NOx 15 emissions by 20 ppm. This derivative is at roughly 1900 K. The impact of residence time is expressed in ppm/ms since this rate is function of the flame temperature.

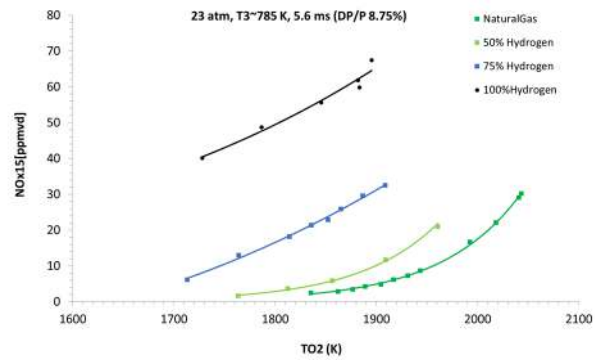

Figure 14. Impact of hydrogen addition on NOx15 emissions at 8.75% DP/P which is equivalent to 5.6ms

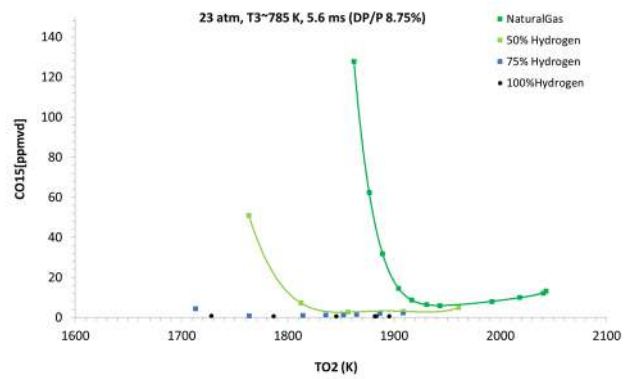

Figure 15. Impact of hydrogen addition on CO15 emissions at 8.75% DP/P which is equivalent to 5.6ms

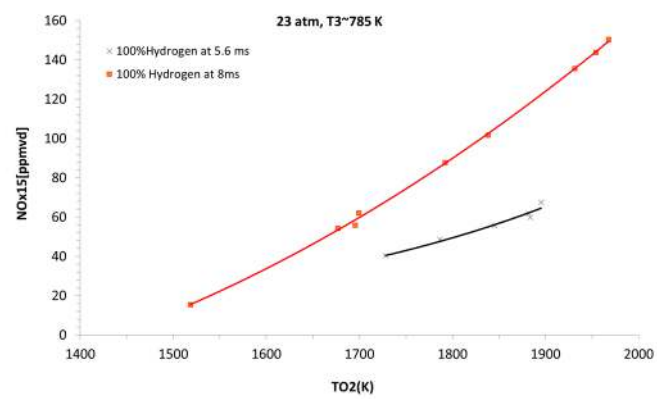

Figure 16. Impact of residence time on NOx15 under 100% hydrogen operation

4.6 NOx vs NOX15

The EPA regulations regulate NOx15 and CO15, rather than NOx and CO. These modified emissions metrics represent the

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respective species emissions corrected to 15% oxygen using the following formula:

$$\boldsymbol{X15} = \frac{\mathbf{20.9-15}}{\mathbf{20.9-O2}\ [\%]} * \boldsymbol{X}\ [\boldsymbol{ppm}] \qquad (3)$$

where X denotes the species concentration in ppm and O2 is the nominal oxygen percentage in the original measurement.

Figure 17 shows the oxygen content in the exhaust for natural gas and pure hydrogen flames (data taken from the PSR-PFR model results discussed in section 4.2). It is evident that natural gas flames have on average 2% less oxygen than respective hydrogen flames. This means that the actual impact on NOx is lower than the reported impact on NOx15. Figure 18 shows the impact of hydrogen addition on NOx vs NOX15. It is approximately 50% of the impact we discussed throughout the paper. Therefore, therefore authors suggest that the emissions regulations should be based on NOx and not NOx15 for combustors burning 100% hydrogen. Eventually, NOx is the final pollutant that is released to atmosphere. In this case, correcting to 15% Oxygen actually penalizes the system with no benefit.

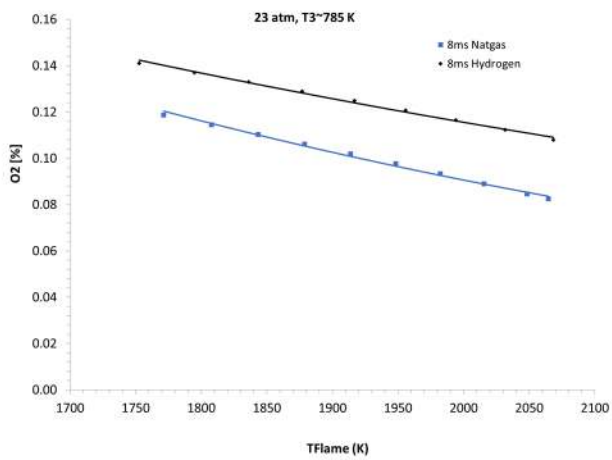

Figure 17. Oxygen percentage for natural gas vs hydrogen flames under same TFlame.

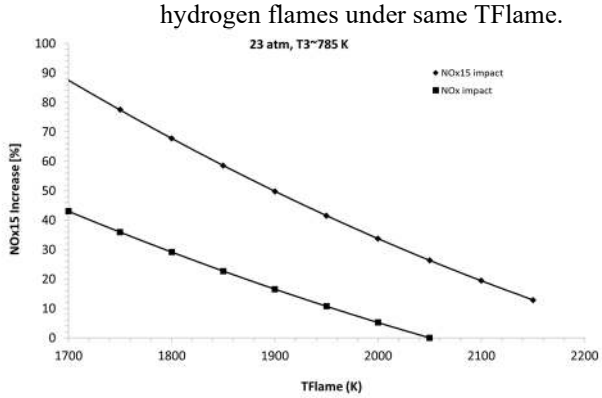

Figure 18. Impact of hydrogen addition on NOx and NOX15

5 CONCLUSIONS

Reactor network modelling and experimental data are used to show the challenge that associates hydrogen combustion up to 100%. Modelling shows that interms of flash back, DLE systems offers advantages over RQL combustion systems as the flame speed gores significantly with higher equivalence ratio until the ratio to natural gas peaks at around 50:1. Also, Blow Off Time (BOT) shows that the problem of burning hydrogen can be classified into three regions. First region is up to 25% hydrogen where the BOT is very similar to propane. Second region is 25%-80% where the BOT significantly drops lower than propane. Finally, more than 80% where the propane BOT levels at 3.5 micros seconds but the hydrogen BOT continues to drop to lower than 1 micro seconds. Therefore, the DLE system should be design with features that can enable high hydrogen without flame holding. The model predicts a penalty in emissions that decreases with the increase in flame temperature. It ranges between 65-40% within the temperatures of interest. Experimental data on an advanced mixer shows even higher penalty. A proposed frame work shows that a drop in mixedness by ~6% can explain this additional increase in NOx. That is the mixer was used as is without restoring the jet momentum ratio back to the design point under hydrogen operation (the mixer is designed to burn natural gas). The system residence time was dropped from 8 ms to 5.5 ms and therefore more than 60% drop in NOx emissions was achieved. The data presented here shows that the future direction for combustors that will enable high hydrogen operation is compactness and short residence time for true near zero emissions or at least to maintain the level of NOx similar to natural gas operation.

6 REFRENCES

[1] Mohammad, B. S., McManus, K., Brand, A., Elkady, A. M., and Cuppoletti, D., "Hydrogen Enrichment Impact on Gas Turbine Combustion Characteristics," Proc. ASME Turbo Expo 2020: Turbomachinery Technical Conference and ExpositionV04BT04A005.

[2] York, W. D., Ziminsky, W. S., and Yilmaz, E., 2013, "Development and Testing of a Low NOx Hydrogen Combustion System for Heavy-Duty Gas Turbines," Journal of Engineering for Gas Turbines and Power, 135(2).

[3] Lückerath, R., Meier, W., and Aigner, M., 2008, "FLOX® Combustion at High Pressure With Different Fuel Compositions," Journal of Engineering for Gas Turbines and Power, 130(1).

[4] Lantz, A., Collin, R., Aldén, M., Lindholm, A., Larfeldt, J., and Lörstad, D., 2014, "Investigation of Hydrogen Enriched Natural Gas Flames in a SGT-700/800 Burner Using OH PLIF and Chemiluminescence Imaging," Journal of Engineering for Gas Turbines and Power, 137(3).

[5] Kim, H., Arghode, V., and Gupta, A., 2009, "Flame characteristics of hydrogen-enriched methane–air premixed swirling flames," International Journal of Hydrogen Energy, 34(2), pp. 1063-1073.

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[6] Therkelsen, P., Werts, T., McDonell, V., and Samuelsen, S., 2009, "Analysis of NOx Formation in a Hydrogen-Fueled Gas Turbine Engine," Journal of Engineering for Gas Turbines and Power, 131(3).
[7] Griebel, P., Boschek, E., and Jansohn, P., 2007, "Lean Blowout Limits and NOx Emissions of Turbulent, Lean Premixed, Hydrogen-Enriched Methane/Air Flames at High Pressure," Journal of Engineering for Gas Turbines and Power, 129(2), p. 404.
[8] Cozzi, F., and Coghe, A., 2006, "Behavior of hydrogen-enriched non-premixed swirled natural gas flames," International Journal of Hydrogen Energy, 31(6), pp. 669-677.
[9] Leonard, G., and Correa, S., "NOx Formation in Premixed High-Pressure Lean Methane Flames," Proc. Fossil Fuel Combustion Symposium: 13th Annual Energy-Sources Technology Conference and Exhibition, pp. 69-74.
[10] Bhargava, A., Kendrick, D. W., Colket, M. B., Sowa, W. A., Casleton, K. H., and Maloney, D. J., "Pressure Effect on NOx and CO Emissions in Industrial Gas Turbines," Proc. ASME Turbo Expo 2000: Power for Land, Sea, and AirV002T02A017.
[11] Rutar, T., and Malte, P. C., 2002, "NOx Formation in High-Pressure Jet-Stirred Reactors With Significance to Lean-Premixed Combustion Turbines," Journal of Engineering for Gas Turbines and Power, 124(4), p. 776.
[12] Biagioli, F., and Guthe, F., 2007, "Effect of pressure and fuel–air unmixedness on NOx emissions from industrial gas turbine burners," Combustion and Flame, 151(1-2), pp. 274-288.
[13] Rutar, T., Horning, D. C., Lee, J. C. Y., and Malte, P. C., "NOx Dependency on Residence Time and Inlet Temperature for Lean-Premixed Combustion in Jet-Stirred Reactors," Proc. ASME 1998 International Gas Turbine and Aeroengine Congress and ExhibitionV003T06A036.
[14] Elkady, A. M., Herbon, J., Kalitan, D. M., Leonard, G., Akula, R., Karim, H., and Hadley, M., 2012, "Gas Turbine Emission Characteristics in Perfectly Premixed Combustion," Journal of Engineering for Gas Turbines and Power, 134(6).
[15] Mohammad, B. S., Balasubramanyam, P., McManus, K., Ruszczyk, J., Elkady, A. M., and Mueller, M. A., "Combustion Dynamics Diagnostics and Mitigation on a Prototype Gas Turbine Combustor," Proc. ASME Turbo Expo 2012: Turbine Technical Conference and Exposition, pp. 383-389.
[16] Elkady, A. M., Brand, A. R., Vandervort, C. L., and Evulet, A. T., "Exhaust Gas Recirculation Performance in Dry Low Emissions Combustors," Proc. ASME 2011 Turbo Expo: Turbine Technical Conference and Exposition, American Society of Mechanical Engineers, pp. 1183-1192.
[17] Graham, G., Zhuyin, R., Hendrik, F., Liuyan, L., Venkat, T., and Hasan, K., "Modeling CO With Flamelet-Generated Manifolds: Part 2 — Application," Proc. ASME Turbo Expo 2012: Turbine Technical Conference and Exposition, American Society of Mechanical Engineers, pp. 1177-1184.
[18] Karim, G. G., Zhuyin, R., Hendrik, F., Liuyan, L., Venkat, T., and Hasan, "Modeling CO With Flamelet-Generated Manifolds: Part 1—Flamelet Configuration," Proc. ASME Turbo Expo 2012: Turbine Technical Conference and Exposition, American Society of Mechanical Engineers, pp. 1141-1151.
[19] Mohammad, B. S., Volk, B., and McManus, K., "Impact of Secondary Air Flow on Combustor Emissions," Proc. ASME Turbo Expo 2020: Turbomachinery Technical Conference and ExpositionV04BT04A007.
[20] Agency, E. P., "Method 7e," https://www.epa.gov/sites/production/files/2017-08/documents/method_7e.pdf.
[21] Kerstein, A. R., and Ashurst, W. T., 1992, "Propagation rate of growing interfaces in stirred fluids," Physical Review Letters, 68(7), pp. 934-937.
[22] Blouch, J., Li, H., Mueller, M., and Hook, R., 2012, "Fuel Flexibility in LM2500 and LM6000 Dry Low Emission Engines," Journal of Engineering for Gas Turbines and Power, 134(5).
[23] Elkady, A. M., Kalitan, D. M., Herbon, J., Leonard, G., Akula, R., Karim, H., and Hadley, M., 2012, "Gas Turbine Emission Characteristics In Perfectly Premixed Combustion," Journal of Engineering for Gas Turbines and Power, 134(6).

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